

Compiler-Validated Semantic Editing for Weak Local Coding Models

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Abstract

Coding agents edit source code as text, and small locally runnable models fail at it: diff formats break, identifiers get guessed, and errors surface only after files change. ago is a semantic edit protocol for Go. The agent queries a typechecked workspace and submits mutations from a versioned operation catalog; the compiler validates every mutation against an in-memory snapshot before anything reaches disk. Edits apply as atomic, generation-checked transactions or not at all, so an edit that does not typecheck is unrepresentable, not merely checked. Rejections are the conversation: each carries typed diagnostics and paste-ready repairs (the corrected call, verbatim), and resending a rejected call unchanged escalates. An oracle replays each task’s ground-truth commit through the protocol before any model attempts it, so a task enters the benchmark only once it is proven solvable; tasks are mined from real commits in traefik, vault, boundary, and cobra. Across 176 recorded episodes, the semantic mode passes 75% versus 55% for raw file editing under identical prompts and budgets, and the gap widens as models shrink: a quantized 9B model passes 75% of episodes through the protocol against 45% raw, with raw at zero on five of eleven tasks the protocol completes. All code, per-episode evidence (transcripts, request logs, serving configurations), and analysis tooling are released.

1 Introduction

TODO(guy): expand from idea.md.

2 Related Work

Symbol-level agent tooling over LSP (Serena; gopls’s experimental MCP server) navigates semantically but still edits text, or exposes a single mutation. The compiler diagnostics sit right there in those stacks; the difference here is that validation is not advisory. CODESTRUCT frames agents over structured action spaces with syntax-validated edits and reports token and accuracy gains; ago’s mutations are type-checked, transactional, and carry repairs. Editing-format benchmarks (EDIT-Bench, RES-Q, FEA-Bench) measure text formats and find that strict formats hurt current models. That is the failure mode the repair channel converts: the format stays strict and the rejection hands back the corrected call. Compiler-guided patch generation (Repilot) and typed-hole context (Hazel) take the same validate-during-generation stance at expression granularity. TODO: full citations.

3 The ago Protocol

3.1 Surface

Ten tools carry a versioned catalog of 33 operations across declaration, statement, test, and project families, plus seven query kinds with deterministic, paged responses. Node handles address statements inside declarations; generation counters make staleness explicit instead of silent.

3.2 Transactions

An ordered op list validates as one unit against an in-memory snapshot. Ops apply to a copy, the dirty set re-typechecks once (in parallel, scheduled over post-edit import edges), resolution proofs run (rename rejects reference capture the compiler would happily accept), then all files write together or nothing does. Pre-existing issues in the affected packages are baselined and tolerated; real codebases carry history, and only new diagnostics reject.

3.3 Rejection as conversation

Every rejection answers one question: what is the correct next call. It carries typed diagnostics, candidate lists, and `possible_repairs`, complete calls the agent resends verbatim. Repairs are tested by execution before they are offered; a repair that would itself reject is worse than none. Resending a rejected call unchanged escalates. Accepted mutations embed the fresh view of the touched declaration, so back-to-back edits skip the read round trip.

4 Oracle Methodology

Every benchmark task comes from a real commit, so the correct answer is already known. Before any model attempts a task, the oracle submits that known answer through the protocol itself. If the right answer cannot get through, the model never had a chance: the problem is ours (an engine bug, an extractor error, or a named protocol ceiling), and the task stays out of the roster until it is fixed. This separates “the model failed” from “the task was impossible” by construction, and the oracle transcripts double as a training corpus.

5 Benchmark

Same model, same harness, same prompts (token parity is enforced by test), same time cap. One arm edits files with shell tools; the other sees only the protocol. Scoring is a goal predicate plus a clean typecheck plus scoped tests, and the test gate is voided where a pristine worktree also fails it. Serving configurations are pinned per run: GLM-4.7-Flash (MoE) at UD-Q4_K_XL, roughly 20 GiB with a q8_0 K/V cache at 128k context; gpt-oss-20b at MXFP4, 11.28 GiB of weights before KV; Qwen3.5-9B at Q4_K_M, 5.28 GiB, deliberately not the larger UD quant because its 8 GB card also hosts an embedding server. Local serving is part of the experiment, constraints included. Every episode records its transcript, diff, score, token usage, failure kind, and the daemon’s request log; the evidence is the contribution as much as the numbers.

6 Results

Semantic passed 66/88 episodes; raw passed 48/88. The per-task table tells the sharper story: raw wins only where grep suffices, and goes to zero on the tasks that need whole-workspace consistency. TODO: repair-uptake analysis (the 9B model recovers through rejections it is offered no repairs for; the 30B-class model ignores repairs it is offered), ablation arm, floor-model section, the cff8d42 case study (discovery-primitive gap found and fixed).

7 Limitations

One language. Contamination is held constant across arms but inflates absolute rates. One serving stack per model, small k, and the infrastructure is author-run. TODO: expand honestly.

8 Availability

Code (MIT), all episode evidence, and analysis tooling: <https://github.com/guygrigsby/agent-go>.

profile	task	mode	pass@k	median green (s)	capped	resends
glm-flash	boundary_06d69f69	raw	3/4	394	0	0
glm-flash	boundary_06d69f69	semantic	4/4	250	0	0
glm-flash	boundary_35b91c66	raw	1/4	720	4	0
glm-flash	boundary_35b91c66	semantic	2/4	643	1	0
glm-flash	boundary_9237d6f7	raw	0/4	–	4	0
glm-flash	boundary_9237d6f7	semantic	1/4	244	0	0
glm-flash	boundary_9f2c83f3	raw	2/4	720	4	0
glm-flash	boundary_9f2c83f3	semantic	4/4	377	0	0
glm-flash	boundary_c0532428	raw	4/4	425	0	0
glm-flash	boundary_c0532428	semantic	3/4	336	2	0
glm-flash	boundary_caf19f86	raw	2/4	615	1	0
glm-flash	boundary_caf19f86	semantic	2/4	357	2	0
glm-flash	cobra_0960ff7f	raw	4/4	690	2	0
glm-flash	cobra_0960ff7f	semantic	4/4	720	3	0
glm-flash	traefik_51491463	raw	2/4	531	2	0
glm-flash	traefik_51491463	semantic	4/4	464	1	0
glm-flash	vault_95b24f55	raw	4/4	260	0	0
glm-flash	vault_95b24f55	semantic	4/4	292	0	0
glm-flash	vault_caec65a7	raw	2/4	720	4	0
glm-flash	vault_caec65a7	semantic	4/4	393	0	0
glm-flash	vault_cfff8d42	raw	4/4	306	0	0
glm-flash	vault_cfff8d42	semantic	0/4	–	4	0
qwen3.5-9b	boundary_06d69f69	raw	3/4	71	0	0
qwen3.5-9b	boundary_06d69f69	semantic	4/4	167	0	0
qwen3.5-9b	boundary_35b91c66	raw	0/4	–	0	0
qwen3.5-9b	boundary_35b91c66	semantic	3/4	162	0	9
qwen3.5-9b	boundary_9237d6f7	raw	0/4	–	0	0
qwen3.5-9b	boundary_9237d6f7	semantic	2/4	615	0	6
qwen3.5-9b	boundary_9f2c83f3	raw	0/4	–	0	0
qwen3.5-9b	boundary_9f2c83f3	semantic	2/4	138	0	0
qwen3.5-9b	boundary_c0532428	raw	3/4	72	0	0
qwen3.5-9b	boundary_c0532428	semantic	2/4	68	0	10
qwen3.5-9b	boundary_caf19f86	raw	0/4	–	0	0
qwen3.5-9b	boundary_caf19f86	semantic	3/4	139	0	1
qwen3.5-9b	cobra_0960ff7f	raw	4/4	165	0	0
qwen3.5-9b	cobra_0960ff7f	semantic	4/4	156	0	86
qwen3.5-9b	traefik_51491463	raw	3/4	57	0	0
qwen3.5-9b	traefik_51491463	semantic	4/4	114	0	0
qwen3.5-9b	vault_95b24f55	raw	4/4	126	0	0
qwen3.5-9b	vault_95b24f55	semantic	4/4	254	0	0
qwen3.5-9b	vault_caec65a7	raw	0/4	–	0	0
qwen3.5-9b	vault_caec65a7	semantic	4/4	73	0	0
qwen3.5-9b	vault_cfff8d42	raw	3/4	64	0	0
qwen3.5-9b	vault_cfff8d42	semantic	2/4	73	1	8

Table 1: Per-task, per-mode outcomes. Generated from committed episode evidence; pass@k over k episodes, median time-to-green over passing episodes.